

SURENDRA KUMAR MISHRA

Head of Department – Polymerization | Chemical Engineer | 18+ Years of Leadership Experience

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Ayutthaya, Thailand | Open to Global Opportunities

CAREER OBJECTIVE

Performance-driven Chemical Engineer with over 18 years of progressive leadership in polyester and petrochemical manufacturing. Specialist in PET chips production, continuous polymerization, SSP operations, and large-scale plant management with capacities up to 1,800 TPD. Adept at process optimization, strategic cost reduction, and operational excellence. Seeking a senior leadership role — Plant Head | AVP | GM Operations | Director Manufacturing — to drive productivity, quality, and profitability across world-class manufacturing facilities.

PROFESSIONAL SUMMARY

- 18+ years of expertise in PET chips, polymerization, SSP, and petrochemical plant operations.
- Proven track record in troubleshooting, continuous improvement, and end-to-end project leadership from planning to commissioning.
- Expert in large-scale continuous process plant management: production planning, quality control, maintenance coordination, and manpower optimization.
- Hands-on Greenfield and Brownfield project execution with up to 40% capacity enhancement achieved.
- Proficient in advanced DCS platforms: DeltaV, Fisher Rosemount, Yokogawa Centum CS-3000 & XL, Supcon, Siemens PCS-7, Novatech Belgium.
- P&L-driven leadership delivering measurable cost savings across raw materials, energy, and manpower.
- Strong HSE leadership — Process Hazard Analysis (PHA), HITRA, permit-to-work systems, Emergency Response (ERCP).
- Certified Six Sigma Green Belt (IIT Delhi), Lean Project Management & Lean Fundamentals Certified (LeanPM.org, 2025).
- Managed cross-cultural teams of 150+ personnel across India and Thailand.

PROFESSIONAL EXPERIENCE

Sunflag (Thailand) Ltd. | *Ayutthaya, Thailand*

Head of Department – Polymerization *Jul 2025 – Present*

- Lead integrated operations of a 290 TPD CP & PSF plant producing POY, FDY, HOY, SD Chips, and Polyester Staple Fiber (PSF).
  - CP Section (Zimmer, Germany): Design 120 TPD → Actual 150–180 TPD (25–50% above nameplate capacity)
  - PSF Section (Tonghui, China): Design 100 TPD → Actual 110 TPD
- Drive operational excellence through rigorous process optimization, equipment reliability, and effective manpower deployment.
- Achieved measurable energy cost reduction per ton through fuel consumption analysis and process parameter tuning.
- Improved plant stability by minimizing POY/FDY breaks per ton, enhancing OEE above 95%.
- Strengthened quality, safety, and statutory compliance; led continuous improvement initiatives across all production lines.
- Developed workforce capability through structured technical training, knowledge-sharing, and lean methodology programs.

Indorama Synthetic India Ltd. | *Butibori, Maharashtra, India*

Assistant General Manager – CP & SSP *Jul 2024 – Jun 2025*

- Directed operations of three Continuous Polymerization (CP) lines (900+ TPD) and one SSP plant (600 TPD) using Chemtex & Polymetrix technologies.
- Led plant-wide performance optimization, yield improvement, and variability reduction initiatives.
- Spearheaded CAPEX projects, shutdown planning, and commissioning ensuring safe, on-time, cost-effective execution.
- Managed major equipment: esterifiers, UFPP/DRR reactors, distillation columns (packed & bubble cap), ejectors, LRVPs, heat exchangers, dryers, compressors, pumps.
- Strengthened OMS, IMS, and FSMS frameworks for improved process governance and audit readiness.
- Drove root cause analyses, SOP enhancements, P&ID/PFD updates, and reliability-centred maintenance practices.
- Reinforced process safety, environmental compliance, and strong HSE culture across operations.

Reliance Industries Ltd. | *Dahej, Gujarat, India*

Assistant General Manager – PET-3 Plant *May 2018 – Jun 2024*

- Key leader for 1,800+ TPD PET complex (two CP & SSP lines: CHEMTEX CP / BUHLER SSP), ensuring high plant availability and product quality excellence.
- Delivered energy cost savings of ₹3 Crore per annum through Dowtherm furnace (Ethane & LNG based) optimization, heat integration, and utility improvements.
- Led Process Hazard Analysis (PHA), HITRA studies, permit-to-work systems, and Emergency Response (ERCP) training as Plant HSEF Coordinator.
- Managed OSBL including heavy equipment (pumps, compressors, chillers, furnaces, chlorine system) start-up, changeover, and rotary equipment performance monitoring.
- Handled filter cleaning systems (hydrolysis, hydro-jetting, and oxidation cleaning processes).
- Implemented Operating Management System (OMS), Production Accounting System (PAS), and reviewed/updated SOPs through Job Cycle Checks.
- Participated in daily rake-up meetings, monthly reliability meetings, PSAT & FPC reviews, and People's Council meetings.
- Mentored shift engineers and frontline leaders; conducted ERCP scenario training for all FEs and POs.
- Certified Internal Auditor – FSMS ISO 22000:2018 & IMS (DNV); awarded Best Plant Safety Representative (2021–2023) and Best HSEF Coordinator (2022–2023).

Wellknown Polyester Ltd. | *Daman (UT), India*

Assistant Manager – Continuous Polymerization (CP-1) *Jan 2011 – Apr 2018*

- Managed 1×300 TPD + 3×200 TPD CP plants with integrated POY/FDY units producing super bright, semi-dull, full dull, and cationic dyeable grades.
- Successfully commissioned 3×200 TPD CP plants — Huitong, Oerlikon & Barmag technologies (China); all three units (Homo Bright, CD Chips, Full Dull) erected, commissioned, and started up.
- Prepared TiO_2, SiO_2 slurry, and $BaSO_4$ through dry mills, centrifuges, and related equipment for specialty chip grades.
- Achieved fuel savings through energy audits and conversion from liquid to gas-based fuels; reduced POY/FDY breaks by 15% through parameter optimization.
- Modified chipper dryer mesh design, eliminating long strand customer complaints and reducing classifier oversize/undersize chip losses.
- Prepared and implemented ISO 9001:2008 quality systems, SOPs, and safety standards across operations.
- Served as Plant Safety Representative (2 years) and Shutdown Team Leader (3 major shutdowns); led Quality Circle and 5S implementation teams.

Jindal Poly Films Ltd. | *Nashik, Maharashtra, India*

Shift Engineer – Poly Plant *Apr 2009 – Jan 2011*

- Operated 320 TPD batch process and 150 TPD CP plant (Polytex Technology, China) producing bright, silica, milky white, opaque white, and matte grades.
- Supported erection, commissioning, and startup of CP unit; operated Yokogawa DCS for process control.

Uflex Ltd. | *Bhind, Madhya Pradesh, India*

Shift Engineer – Polymerization *Aug 2007 – Mar 2009*

- Joined as Graduate Trainee Engineer; promoted within one year to Shift Engineer managing 186 TPD batch process PET plant.
- Operated Yokogawa DCS for process control; produced bright, silica, and clear transparent PET grades.
- Underwent industrial training at J.K. Tyres Industries Ltd., Banmore, Gwalior (MP).

KEY PROJECTS & PROCESS IMPROVEMENTS

- Catalyst Changeover (ATO → ATG → ATA): Led strategic catalyst transition reducing antimony-based catalyst dependency, cutting per-ton catalyst cost, eliminating supply chain risk, and achieving compliance with global food-contact and environmental quality standards.
- Ti Co-Catalyst Trial: Pioneered titanium-based co-catalyst trials as a sustainable alternative to antimony, targeting significant cost reduction and positioning the plant ahead of tightening global environmental regulations.
- Hot Well Glycol Overflow Modification: Designed and implemented system modification simultaneously reducing EG column heat load, suppressing EG degradation byproducts, improving polymer colour tone (b* value), and delivering a measurable 0.3 kg/ton fuel oil saving — a recurring annual cost benefit.
- Manpower Optimization: Implemented multi-skill cross-training across 150+ plant personnel, increasing operational flexibility, reducing shift dependency, and cutting overtime costs by 20% — directly improving plant OPEX.
- Chipper Dryer Mesh Redesign (Wellknown): Modified mesh design to eliminate long strand defects and reduce classifier oversize/undersize chip losses — improved customer satisfaction and reduced product rejections.
- Energy Optimisation (Reliance): Furnace and utility system optimization delivering ₹3 Crore annual energy savings.
- POY/FDY Break Reduction (Wellknown): Systematic process parameter optimization reducing product breaks and rejection rates by 15%.

EDUCATION

Bachelor of Engineering – Chemical Engineering *Jiwaji University (IEJU), Gwalior (MP), India*	*66% | 2003–2007*
Higher Secondary (12th) – Vivekananda HSC, Shahdol, MP	*65.2% | 2003*
Secondary (10th) – Govt. HSC, Malanga, MP	*60.2% | 2001*

CERTIFICATIONS & PROFESSIONAL RECOGNITION

✦ Six Sigma Green Belt	*IIT Delhi – October 2014*
✦ Lean Project Management Certified	*LeanPM.org – 2025*
✦ Lean Fundamentals Certified	*LeanPM.org – 2025*
✦ Internal Auditor – FSMS ISO 22000:2018	*DNV Certified*
✦ Internal Auditor – IMS	*DNV Certified*

✦ Best Plant Safety Representative	*Reliance Industries – 2021–2022 & 2022–2023*
✦ Best Plant HSEF Coordinator	*Reliance Industries – 2022–2023*
✦ Site-Level PMR (Process Management Representative)	*Reliance Industries Ltd.*

TECHNICAL SKILLS

DCS Systems	DeltaV, Fisher Rosemount, Yokogawa Centum CS-3000 & XL, Supcon, Siemens PCS-7, Novatech Belgium
Software	MS Office, MS Excel, OMPro, SAP (Production & Planning Module)
Engineering	P&ID, PFD, GA, ISO drawings; Process Hazard Analysis (PHA); HITRA; OMS; FSMS; IMS
Equipment	Reactors, Esterifiers, UFPP, DRR, Distillation Columns, Ejectors, LRVP, Heat Exchangers, Furnaces, Compressors, Centrifugal/Screw/Gear/Canned Motor Pumps
Languages	English (Fluent), Hindi (Native)

PERSONAL DETAILS

Date of Birth: 02 July 1984	**Gender:** Male
Marital Status: Married	**Nationality:** Indian
Father's Name: Mr. R. N. Mishra	**Permanent Address:** Vill-Amadand, Post Malaga, Tehsil-Kotma, Dist-Anuppur, MP – 484446

REFERENCES

Mr. Sanjay Mhaske – GM Operations, Garware Hi-Tech Films Ltd | +91 77091 33611
Mr. Anish Dutta – HOD, Dhanashree Polymers Ltd | +91 70165 03671
Mr. Siva Kumar – Former HOS, PET-3, Reliance Industries Ltd | +91 97905 52407
Mr. R. H. Shukla – AVP, Wellknown Polyester Ltd | +91 99099 90743

Declaration:

I hereby declare that all information provided above is true and correct to the best of my knowledge and belief.

Place: Ayutthaya, Thailand
Date: _____

Surendra Kumar Mishra
Head of Department – Polymerization
Sunflag (Thailand) Ltd.